

Midterm Exam Format

This is what you can expect the format of the exam to be like. There will be 13 questions and this document tells you exactly the topics of each question so that you can prepare accordingly.

1. Algorithm analysis

- Function runtime analysis (tight bound using Θ)
- Recursive functions
- Understanding the runtime for:
 - i. Discussed data structures
 - ii. Discussed algorithms
- STL sequence container operations

2. Algorithm analysis continued

- Use of iterators
- Amortized analysis
- Space complexity analysis

3. Recursion

- Time-complexity
- Recurrence using functional notation

4. Sorting

- Tracing a given array for:
 - i. Insertion Sort
 - ii. Merge Sort
 - iii. Quick Sort
 - iv. Heap Sort
- Understanding of quick sort pivot selection
- Understanding of merge process in merge sort

5. Big-Five:

- Understanding the code declaration of each:
 - i. Destructor
 - ii. Copy constructor
 - iii. Copy assignment
 - iv. Move constructor
 - v. Move assignment

6. Understanding of constructor behavior in C++

7. Understanding of C++11 enhancements for big-five:
 - i. Default behavior
 - ii. Move semantics
 - iii. Smart pointers
8. Implementation of the big-five
9. Understanding where the big-five are invoked given some code
10. Binary Trees and Heaps
 - o Min-heap vs max-heap
 - o Binary tree traversal:
 - i. In-order
 - ii. Pre-order
 - iii. Post-order
 - o Adding to a heap: step-by-step process
 - o Removing from a heap: step-by-step process
11. AVL tree insert
 - o Ability to insert to an AVL tree
 - i. BST insert
 - ii. Imbalance detection
 - iii. AVL rotations to restore balance
12. AVL tree remove
 - o Ability to remove from an AVL tree
 - i. BST remove
 - ii. Imbalance detection (can be multiple for remove)
 - iii. AVL rotations to restore balance
13. Implementation (one from the following):
 - o Sorting algorithms we discussed in class
 - o Selection algorithms we discussed in class
 - o Iterative tree traversal using STL stack or queue

Note: You should review all quizzes, lectures, and projects to fully prepare for the exam. The topics mentioned here are not the only things you should review if you want to get a good grade. Review anything that you are not sure about from slides on the course webpage.